

Facultad de Informática / Faculty of Computer Science ???

Trabajo de Fin de Grado / Bachelor's Thesis ???

AtrilMatic: A Web-based Smart Music Stand

Por/By???
Tim Matveev y Elena Juarros
Calificación final individual/Individual final grade: ??



UNIVERSIDAD COMPLUTENSE
MADRID

Dirigido por / Supervised by ???

Manuel Freire Morán

MADRID, 2026–2027

Acknowledgments

Viva el director

Abstract

AtrilMatic: A Web-based Music-Stand

Keywords: Digital music stand, pitch detection, score following, real-time audio processing, MusicXML, WebAssembly, Rust.

Resumen

AtrilMatic, un atril inteligente

Palabras clave: Atril digital, detección de tono, seguimiento de partituras, procesamiento de audio en tiempo real, MusicXML, WebAssembly, Rust.

Contents

Acknowledgments ii

Abstract iii

Resumen iv

Chapter 1 Introduction **2**

1.1 Motivation 2

1.2 Objectives / goals ?? 2

1.3 Planning 2

Chapter 2 Related work **3**

2.1 Introduction 3

2.2 Technological foundations 3

2.3 Existing solutions 3

2.4 Comparison with the proposed solution 5

Chapter 3 Development **6**

3.1 Design 6

3.2 Implementation 6

Chapter 4 Evaluation **7**

Chapter 5 Conclusions **8**

5.1 Conclusions 8

Capítulo 1 Introducción **9**

1.1 Motivación 9

Capítulo 2 Conclusiones **10**

2.1 Conclusiones 10

Bibliography **11**

AppendixA: Evaluation questionnaire **13**

List of Figures

2.1 Tomplay user interface. 3
2.2 Oktav user interface. 4

List of Tables

Chapter 1. Introduction

1.1. Motivation

1.2. Objectives / goals ??

1.2.1. General objective

1.2.2. Specific objectives

1.3. Planning

Chapter 2. Related work

2.1. Introduction

2.2. Technological foundations

2.3. Existing solutions

Before starting with our own app, we felt that a way of gathering... was looking into already existing solutions in the market to see what we liked or not...???

This section reviews 4 applications that propose similar solutions to what we wanted to do. The solutions analysed are: Tomplay, Oktav, Newzik and Enote.

2.3.1. Tomplay

Tomplay is a mobile and desktop app developed by Tomplay.

maybe we can add here screenshot of the UI from the original web

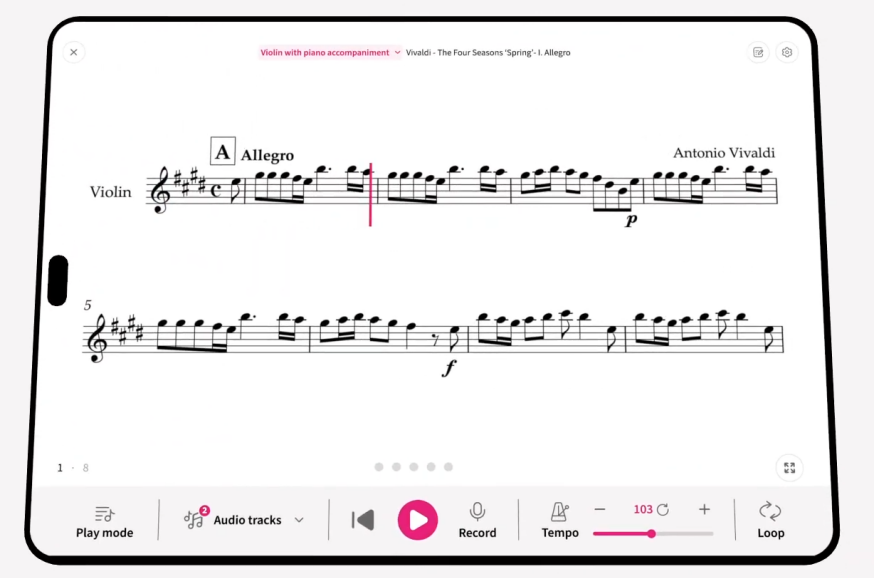


Figure 2.1: Tomplay user interface.

Its main features are:

- **Digital sheet music**: Automatically scrolls in sync with the music.
- **Note by note practice**: The score advances only when you play the right note.
- **Adjust the tempo**: Slow down or speed up the music tempo.
- **Record yourself**: Record and listen back to your performance.
- **Annotate**: Add your own annotations to the music sheet.

- **Big music sheet catalogue**: Over 100,000 scores with professional audio accompaniments.

However, it presents some limitations:

- Paid subscription
-
-

2.3.2. OKTAV

OKTAV is a web app developed by OKTAV.



Figure 2.2: Oktav user interface.

Its main features are:

- **Autoscroll**: Play without the need to turn pages and adjust tempo.
- **Annotate**: Ability to add fingerings, notes or highlight sections.
- **Play-along**: listen and play along to sheet music based on MIDI.
- **Digital piano**: See how to play a music sheet on a visual keyboard shown along the music sheet.

However, it presents some limitations:

- Paid subscription
- Only for piano, violin and cello
-

2.4. Comparison with the proposed solution

Chapter 3. Development

3.0.1. Functional requirements

3.0.2. Non-functional requirements

3.1. Design

3.1.1. System architecture overview

3.1.2. Database design – we dont have DB!

3.2. Implementation

3.2.1. Backend and persistence

Chapter 4. Evaluation

4.0.1. Objectives / goals ??

4.0.2. Participants

4.0.3. Test environment

4.0.4. Evaluation tasks

4.0.5. Procedure

Chapter 5. Conclusions

5.1. Conclusions

5.1.1. Fulfilment of the objectives / goals ??

Capítulo 1. Introducción

1.1. Motivación

1.1.1. Objetivo general

1.1.2. Objetivos específicos

Capítulo 2. Conclusiones

2.1. Conclusiones

Bibliography

AppendixA
Evaluation questionnaire

This appendix presents ... Eg: SUS survey